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RESEARCH ARTICLE | MARCH 22 2024

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Shahnawaz Ahmad ✉; Shabana Mehfuz; Javed Beg

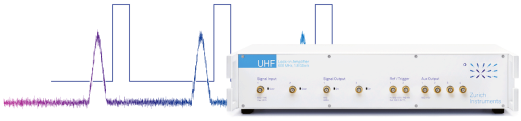
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AIP Conf. Proc. 2816, 150003 (2024)
https://doi.org/10.1063/5.0177484

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Service Centric Attribute Based Block Level Two Step Encryption Scheme for Improved Cloud Security

Shahnawaz Ahmad[1, a)], Shabana Mehfuz[1, b)], and Javed Beg[2, c)]

[1]*Department of Electrical Engineering, Jamia Millia Islamia (A Central University), New Delhi, India.*
[2]*Oracle, Noida, Uttar Pradesh, India*

Corresponding author Email: [a)]*shahnawaz98976@gmail.com*
[b)]*smehfuz@jmi.ac.in*
[c)]*javed.beg@oracle.com*

Abstract. Data security has been identified as key challenge in data maintenance in cloud. Towards maintaining data security, there exist several approaches, but failed to achieve better performance in data security. An efficient service centric attribute-based block level two step encryption scheme (SABTES) is presented in this article to improve the data security performance. The method enforces data security in two ways; first a Service Level Access Restriction (SLAR) can be perfromed according to the user profile being maintained, second; the method enforces block level attribute-based encryption in two steps according to the service characteristics. To perform access restriction, the method computes Legitimate Service Frequency (LSF) measure and Other Legitimate Service Frequency (OLSF). Using these one can perform access restrictions. The approach chooses the encryption scheme and key based on the nature of the service to ensure data security. Similarly, the attributes of the data are encrypted with specific key according to distinct attribute where the merged data are split into blocks and encrypted with the service level key selected. The proposed SABTES introduces higher performance in access restriction and data security.

Key words: *Cloud, Cloud Security, Access Restriction, ABE, Block Ciphers, Two Step Scheme, SABTES, LSF, OLSF, SLAR.*

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INTRODUCTION

The modern informative world is greatly runs over service orient architectures. The global users perform their most works through different online services. The organizations also provide various services for their customers and users which can be accessed through various devices. The service orient architectures are become more popular due to this fact. As of the most human accesses various services available in different networks, the traffic towards the service point become higher. Like any network, the service orient environment faces variety of security challenges. The presence of any malicious node would involve in variety of threats. This would degrade the service performance as well as performance of entire system. In the cloud, data security is ensured in two ways. In terms of access restriction, various methods are available. The first method is profile based approach that restrict the user access according to the user profile and the key based approach verifies the user access according to the key provided. Similarly, the access restriction has been enforced by computing different trust measures according to earlier access made by the user. The behavior-based approach monitors the behavior of the users and based on that the trust measures are computed to restrict the malicious access. Similarly, there are a variety of ways that have been considered in the literature. On the other hand, data security is ensured by the use of data encryption. The data encryption has been performed in service level which selects a key and scheme depends on the service. Similarly, in case of attribute-based approach, the method would select different key and scheme for the attribute. At the same time the session-based approaches are used where the selection of key and scheme will differ at different session time. Similarly, there are a variety of ways in the literature for strengthening cloud data security. This article discusses a novel service centric attribute-based block level two step encryption scheme to improve the security performance. The model is focused on selecting an encryption key and scheme according to the service

Advance Computing and Ingenious Technology in Engineering Science
AIP Conf. Proc. 2816, 150003-1–150003-11; https://doi.org/10.1063/5.0177484
Published by AIP Publishing. 978-0-7354-4789-9/$30.00

characteristic. Also, the data has been encrypted in two steps as first at attribute level and second at block level. Also, the method is designed with enforcing the access restriction to improve the quality of model. The method estimates Legitimate service frequency (LSF) towards own class and other class to measure trust weight. Based on these values, the access restriction is performed.

PREVIOUS WORKS

In various studies, a different number of approaches to data security have been discussed in the literature and set of approaches are discussed in detail in this section.

Concerning data security, different approaches were evaluated among them, the HABE scheme is the most popular which is also known as hybrid attribute-based encryption [1]. It generally uses the CPABE algorithm to maintain access to the cloud [2]. Moreover, in banking models, various authors have utilized different data-sharing methods such as SHAI, MD5, and RSA [3]. Based on merging both BRE and ABE encryption, the techniques of proxy encryption and aggregate were combined together to improve data security [4]. Furthermore, the blowfish algorithm and the porter stemming algorithm were also used widely in data encryption [5].

In [6], the authors discussed an eraser code scheme with proxy re-encryption through which data can be shared with encryption. Based on [7], encryption is a process that performs data authentication and data obfuscation on the server. In addition, a brief study has been conducted to measure the homomorphic methods to improve data security in the cloud [8]. Furthermore, in the same field of study, a different data encryption scheme was presented which is a Diffie Helman with AES [9]. In order to improve encryption and improve data security, various multi-level security strategies have been discussed [10]. In [11], the hybrid encryption method was explained to measure data security in clouds.

In [12], the FHE (Fully Homomorphic Encryption) has been discussed. Whereas hybrid encryption was discussed in [13]. These two methods combine the blowfish and RSA approaches. Moreover, in [14], another homomorphic way was discussed which is the cloud-Elgamal. It usually allows computation in data without decryption. The RSA-CPABE method has been also discussed in [15], which can be referred to as the spitting algorithm.

In [16] the PU-ABE was presented involves capturing different attributes and accessing privacy preservation and encryption. Using homomorphic encryption, an enhanced scheme with PSO has been presented [17].

The HADES-IoT was also discussed and presented in [18] to perform data detection and generate resistance without tampering to the system [19]. The author of [20] presents a hybrid of a genetic algorithm and a back propagation neural network that uses the genetic algorithm for feature selection and the BPNN for attack detection. The author of [21] presents a Bi-LSTM technique based on an attention mechanism that learns by sending IDS data to multiple hidden layers. In this manner, intrusion detection is carried out.

The author of [22] presents an ensemble-based machine learning strategy that incorporates Boosted tree, bagged tree, subspace discriminant, and RUSBooted, as well as a voting scheme. To produce a consolidated final prediction, the framework incorporates the voting algorithm. A network intrusion detection system (NIDS) is presented in [23], which monitor the packet flow and uses signature-based detection system to perform anomaly detection. The methods discussed in literature suffer to achieve higher performance in data security.

SERVICE CENTRIC ATTRIBUTE BASED BLOCK LEVEL TWO STEP ENCRYPTION SCHEME (SABTES)

The proposed service centric attribute-based block level two step encryption scheme maintains the records of previous service access. Using the access trace, the method computes Legitimate Service Frequency (LSF) measure and Other Legitimate Service Frequency (OLSF). Similarly, other methods is focused on accessing the requested services, and over the information obtained the method enforce attribute-based encryption and applies block level encryption in two stages. The user has been given encrypted data.

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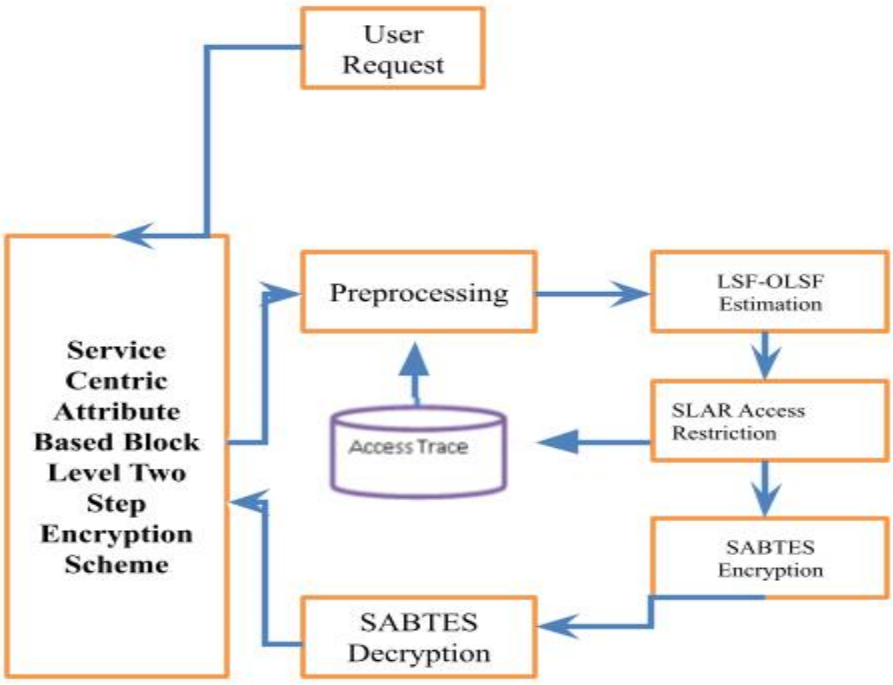


FIGURE 1. The Proposed SABTES Model's Architecture (created by the author)

The functional architecture of the proposed SABTES model is depicted in Fig. 1, and the functional components of the SABTES model are discussed in depth in this section.

Pre-processing

Based on the SABTES model, it maintains service traces access produced through various users of the environment. The service provided by the cloud would be accessed by different users and such traces are used towards access restriction in this model. However, to enforce the access restriction the completeness of the traces is much essential. The approach reads the traces and determines which features are present in the trace. According to the features and their values, method traverses the trace list and finds the completeness of each trace. If there is any trace with incomplete feature and values, then the method eliminates the trace from the list. Such noise removed trace set has been used towards access restriction.

Algorithm

Given: Service Access Trace STrace
Obtain: Noise Removed Trace NRtrace.
Start

Read service access trace STrace.

Identify the set of features Fset = $size(STrace)\ Fset \cup (\sum Features(STrace(i)) \ni Fset)\ i = 1$

For each trace Tk

If Tk$\in(\forall Features(Fset))$ then

if $size(Tk)\ Tk(j) != Null\ j = 1$ then

Add the trace Tk to the preprocessed set.

NRTrace = $\sum(Traces \in NRTrace) \cup Tk$

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End

End

End

Stop

The pre-processing algorithm fetch the service access trace available and finds the set of features present in the trace. Using the feature list identified, the method verifies each trace record for the presence of the feature as well as value. If any of that is missing, then the record is considered as noisy and being eliminated from the set. The noise removed trace has been used in estimating different measures towards access restriction.

LSF-OLSF Estimation

The trust of any user in accessing the services in proper way can be measured by computing LSF and OLSF measures. The legitimacy of any user is depened on the accessing the specific service which can be measured by computing the value of LSF (Legitimate Service Frequency) which defines customer honesty of the user in accessing specific service. Similarly, towards any service, the legitimacy of any user in accessing other service can be measured by computing the value of OLSF (Other Legitimate Service Frequency). The value of LSF is measured by computing the number of times the service being accessed and completed properly with the total number of service access. Also, the value of OLSF is measured by computing total number of other service access and number of times they are completed properly. By using both the values, the access restriction is performed.

Algorithm

Given: Service Access Trace SaT, Service Request Sreq

Obtain: LSF, OLSF

Start

Read SaT, Sreq

Service requested Sr = $Service \in Sreq$

Find the traces of user as UTr = $\sum_{i=1}^{size(SaT)} SaT(i).User == User$

Compute LSF = $\frac{\sum_{i=1}^{size(UTr)} UTr(i).Service==Sr \;\&\&\; UTr(i).State==Finished}{Size(\sum_{i=1}^{size(UTr)} UTr(i).Service==Sr)}$

Compute OLSF = $\frac{\sum_{i=1}^{size(UTr)} UTr(i).Service!=Sr \;\&\&\; UTr(i).State==Finished}{Size(\sum_{i=1}^{size(UTr)} UTr(i).Service!=Sr)}$

Stop

The above discussed algorithm reads the service access traces and finds the user trace. With the user trace available, the method computes the value of LSF and OLSF measures to support access restriction.

SLAR Access Restriction

The proposed SLAR (Service Level Access Restriction) approach reads the request of users and identifies the service which is being requested. With the user identified, the traces belong to the user are split and using them, the method generally calculates the value of LSF and OLSF values. Using the values of both measures, the method helps computing the value of legitimate weight LW to the service that has been restricted.

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SLAR algorithm

Given: Service access Trace SaT, Service Request SReq
Obtain: Null
Starts
Read SaT, Sreq
[LSF, OLSF] = perform LSF-OLSF Estimation.
Compute legitimate weight LW = LSF×OLSF
If LW>Th then
Allow access.
Service Data Sda = Perform service access.
Result R = Perform SABTES Encryption.
Send result to the user.
Else
Deny access.
End
Stop

The algorithm described above illustrates how access limitation and data encryption are carried out based on the user's valid weight. The result of data encryption has been sent to the user.

Data Encryption using SABTES

In two steps, the proposed service centric attribute-based two-step encryption technique operates. The procedure begins by locating a list of features existing in the data. As the method maintains taxonomy of key and scheme set which has information about the key and scheme sets towards various attributes and features, the method chooses a unique key and scheme from the taxonomy. With the scheme and key selected, the method performs data encryption on each feature present in the data. The encrypted data has been merged. In the second stage, the method split large data into smaller data blocks where the number of blocks has been selected in a dynamic passion. Each created block was encrypted with the key and scheme chosen from the service taxonomy. Again, the blocks are merged and produced as result to the user.

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Algorithm

Given: Service Data Sdat, Service Taxonomy ST, Attribute Taxonomy AT
Obtain: Cipher data Cdat, key index set Kis, Scheme index set Sis.
Start

Read Sdat, ST, AT.

Find list of features in service data as Flist = $size(Sdat)\ \sum Features \in Sdat(i)\ i = 1$

For each feature or attribute Ai

Choose the encryption scheme from attribute taxonomy AT.

Scheme s = $Random(AT(, i, size(AT(Ai).Scheme))\ i = 1\ to\ size(AT(Ai))$

Choose the encryption key from attribute taxonomy AT.

Key k = $Random(AT(, i, size(AT(Ai).Key))\ i = 1\ to\ size(AT(Ai))$

Cipher data Cdat(Ai) = Encrypt data feature Ai using key k and scheme s.

Add key k to key index set Kis(Ai) = K.

Add scheme to scheme index set Sis (Ai) = s

End

Encrypted data Ed = Merge the cipher data Cdat

Choose No of blocks Nb = Random (1, size (ST (service). Scheme)))

Divide the data into a number of block Nb.

Block data Bd = Split (Ed,Nb)

Choose the service key Sk = $Random(ST(,i,size(ST(Service).key))\ i\ =\ 1\ to\ size(ST(Service))$

Choose the service scheme Sscheme = $Random(ST(,i,size(ST(Service).Scheme))\ i\ =\ 1\ to\ size(ST(Service))$

Add to key index set and scheme index set.

For each block Bi

Bd (Bi) = Encrypt (Bd (Bi), Sscheme,sk)

End

Cipher data Cdat = Merge blocks and produce result to the user.

Stop

The above-mentioned algorithm demonstrates how to execute two-step attribute-based block level encryption. The method finds the features of service data and for each feature identified, a specific key and scheme are selected to perform data encryption. Once the features of data are encrypted then they are merged to form single one. Furthermore, the approach divides the data into a number of blocks, with the service key and scheme chosen for each block being used to encrypt the data. Encrypted data are merged and produced as result to the user.

SABTES Data Decryption

The proposed SABTES decryption algorithm receives the cipher data with set of key index and scheme index set. The method sends both cipher data and the index set to the user. The index set contains set of index values being used for data encryption and set of scheme index values being used. Also, the method transmits a number specifies the total number of blocks being used. With all these details, the method first split the cipher data into number of blocks according to the number given. Further, for each block of data, the method decrypts the block with service key and merge them. Further, the method finds the number of features and for each feature value, the method select the key and scheme from concern set according to the index value available in the scheme and key index sets. The decrypted results are produced to the user.

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SABTES Decryption

Given: “Cipher Data CD”, “Key set Ks”, “Scheme Set Scs”, “Key index set KiS”, “Scheme Index Set SiS”, “Number N".

Obtain: Original Data ODat.

Start

Read CD, Ks, SCS, KiS, SiS.

Data Blocks Db= Split Data into N number of blocks.

For each block B

Db(B) = Decrypt (Db(B), SCS (SiS (1)), Ks (KIS (1))

End

Data D = merge data blocks.

Feature list Flist = Identify set of features of data D.

For each feature f

F = Decrypt (SCS(SiS(f)), Ks (Kis(f))

End

Result R = Rearrange all features.

Stop

The working of SABTES decryption algorithm has been presented in detail. The method decrypts the data in two stages. In first stage, the block level decryption is performed using service key and in the second stage, the data decryption is performed in attribute level according to the concern key. The decrypted results are produced to the user.

RESULTS AND DISCUSSION

The SABTES algorithm's performance has been implemented and assessed across a number of parameters. The performance evaluation is carried out by considering various test cases. This section presents the detailed analysis of results obtained.

TABLE 1. Evaluation Details

Schemes	Value
Tool Used	Microsoft Azure
No of Services	50
No of Features	100
Number of Users	5000

Table 1 shows the specifics of the setup used for performance evaluation. This section presents and evaluates the performance of the proposed approach.

TABLE 2. Data Security vs. Number of Features Performance Analysis

Schemes	25 Features	50 Features	100 Features
PU_ABE	65	68	72
PSO	69	73	77
HADES_IoT	72	76	83
SABTES	85	89	94

The performance of data security obtained by various ways at various numbers of features in the environment are displayed in Table 2. The suggested SABTES scheme outperforms other PU ABE, PSO, and HADES IoT methods with a security performance of up to 94%.

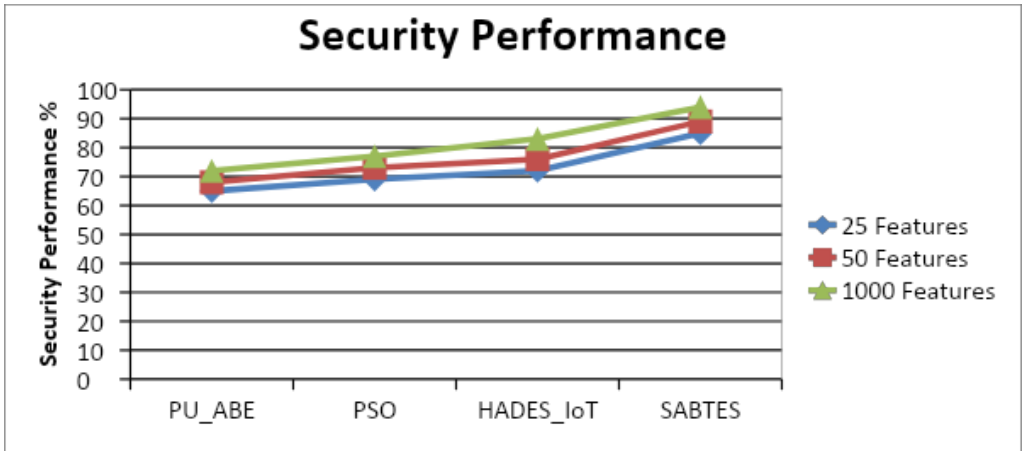


FIGURE 2. Security Performance vs. Users (created by the author)

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TABLE 3. Throughput Performance versus. Number of Features in Percentage

Schemes	25 Features	50 Features	100 Features
PU_ABE	67	69	73
PSO	71	73	75
HADES_IoT	74	77	79
SABTES	84	87	95

Figure. 2 shows the results of the proposed SABTES algorithm in terms of security, as measured by a variety of aspects and compared to the results of other approaches. The SABTES scheme has a security rating of up to 95%, it is quite higher than the current PU ABE, PSO, and HADES IoT systems. In all the cases considered, the performance of security is improved by the proposed SABTES scheme.

Based on various number of features in varying conditions, the throughout performance is evaluated. It has been discussed and explained in the below table. In the table the ratio of throughout produced was presented at each condition. However, it can be identified that the planned SABTES has able to achieve a throughout improvement of 95 percent. It is comparatively higher than other IoT algorithms.

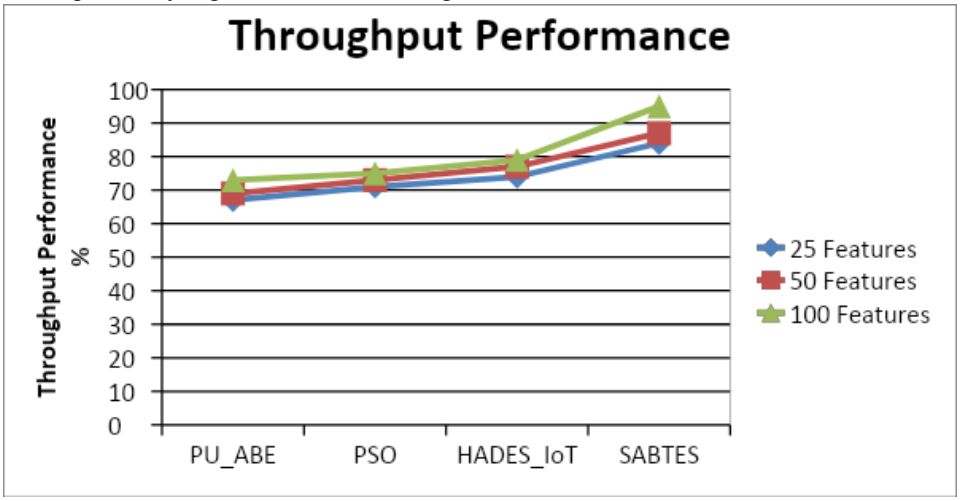


FIGURE 3. Throughput Performance vs Users (created by the author)

Figure 3 shows how throughput performance has been tested in relation to a number of environmental conditions and compared to other methods' results. In comparison to existing approaches, the proposed SABTES algorithms have achieved higher throughput. The suggested SABTES scheme's inclusion helps to improve throughput performance in all of the situations studied.

TABLE 4. Encryption and Decryption Performance vs. number of features

Schemes	25 Features	50 Features	100 Features
PU_ABE	64	67	71
PSO	68	72	76
HADES_IoT	71	75	81
SABTES	86	91	95

Table 4 shows how data encryption and decryption performance is assessed and presented under various settings with several characteristics in the environment. The performance of encryption and decryption is evaluated for each number of user condition. The SABTES system surpassed all other methods in terms of encryption and decryption in all three cases. The proposed solution improved performance by 95%, outperforming other PU ABE, PSO, and HADES IoT methods.

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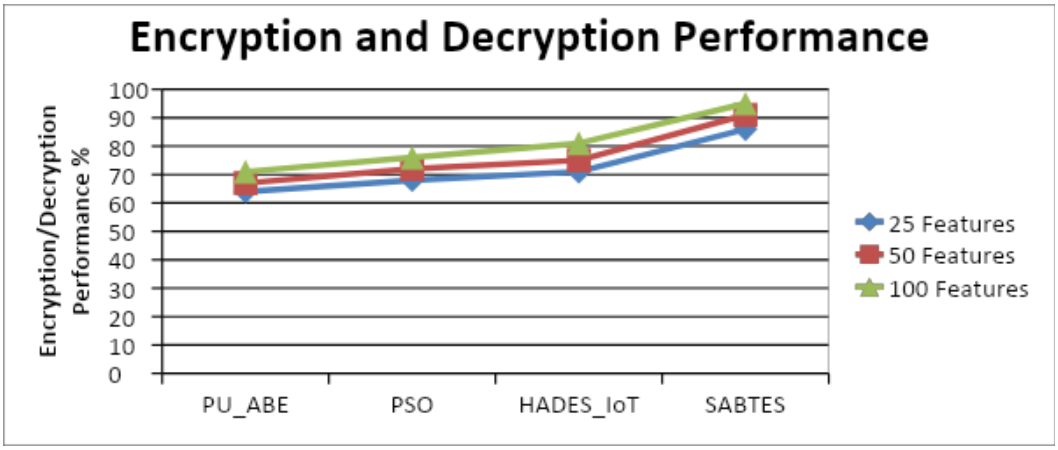


FIGURE 4. Encryption/decryption performance vs. number of features (created by the author)

In Fig. 4, This study has assessed the encryption and decryption performance based on a variety of features and compared with other approaches. Among them, the performance of SABTES algorithms are comparatively higher than other methods. The inclusion of the SABTES scheme supported the performance development more than other methods.

TABLE 5. Time Complexity vs. Number of Features Analysis

Schemes	25 Features	50 Features	100 Features
PU_ABE	85	89	95
PSO	81	85	88
HADES_IoT	70	74	81
SABTES	32	35	43

Table 5 compares and contrasts the time complexity of several approaches for authenticating and restricting user access to a range of functionalities. In comparison to other techniques such as PU ABE, PSO, and HADES IoT schemes, the proposed SABTES algorithm has a lower time complexity.

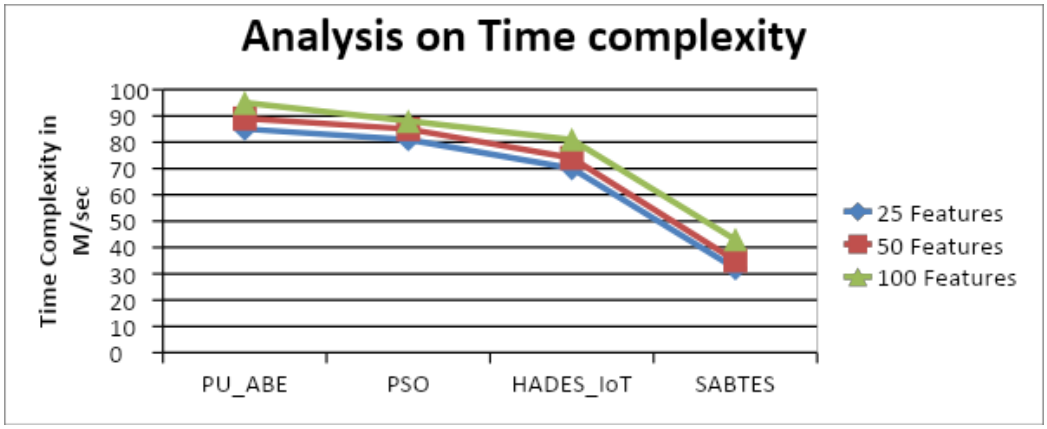


FIGURE 5. Performance in terms of time complexity versus the number of features (created by the author)

In tha above chart the time complexity performance has been measured for a variety of user counts. Other methods have produced more time complexity than the proposed SABTES algorithms. Based on different conditions

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of different features the time complexity performances were evaluated at different conditions. In compared with other methods or approaches, the SABTES method has lesser time complexities [24-26].

CONCLUSION

A service centric attribute-based block level two step encryption (SABTES) strategy for data security in the cloud was provided in this study. The method works on two stages in both encryption and decryption. In first stage, the method performs encryption on attribute level where at the second stage, the service level encryption is performed. At the service level encryption block level approach is used. In the decryption stage, the reverse process is performed. At both the cases, the method uses dissimilar scheme and key for different features which are selected at dynamic manner where the index value of scheme and key pairs are given to the receiver in a set which is being used to conduct the data decryption. The SABTES approach increases the complexity which challenges the adversary and malicious nodes to hide the data. The approach improves the performance of data security and improves the security performance.

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